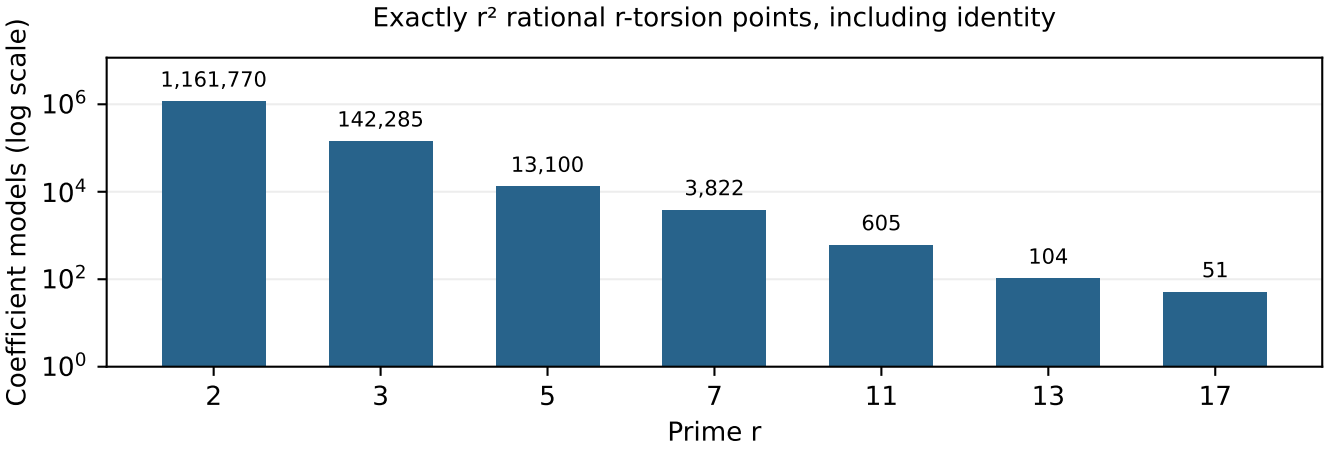
Prime-field BK boundary census below 500

The recorded k=1 census covers all 95 prime fields below 500: all nonsingular general models over F2/F3 and all nonsingular short coefficient pairs for p≥5. Both native passes complete and agree byte-for-byte. Native SQL reconstruction and the canonical export/file check pass. [pcc:unverified]

Exact unit	Count
Coefficient models	7,013,620
Representative/model rows	43,508
Weighted model/prime-factor cases	16,450,388
Qualified prime-to-characteristic k=1 cases	9,668,007
Qualified k=1 mismatches	0
Unrestricted criterion failures	5,386,300
Failures also satisfying r^2 divides #E	2,227,674

There are 43,330 short scaling representatives and 178 general F2/F3 rows. For each prime r dividing #E, the program counts base-field points with rP=O, including O. The unrestricted criterion compares a count of r^2 with r dividing p−1. The qualified subset also requires r≠p and r not dividing p−1; 74,344 characteristic cases are recorded separately. Model weights are not independent random samples. [pcc:unverified]



The figure copies exact native counts; eligibility differs by r. Original BK Theorem1 retains the missing condition, and Remark1 already supplies an r^2-divides-#E counterexample. This census checks the omitted-condition boundary at k=1; extension-field inclusion at all k remains open. [pcc:unverified]

Actual Lean semantics and native assurance

The separate Lean gate passes 29 declarations, zero failures and zero trusted-native exceptions. It constructs the actual additive scaling bijection $(x,y)\to(u^2x,u^3y)$, including tangent/chord/vertical cases and an inverse, before reusing representative counts. [pcc:unverified]

Certificate in BKBasefieldScalingAudit	Checked scope
scaling_audit_certificate	Equal actual point counts and killed-by-n counts for every n
rational_low_torsion_certificate	Actual rational 2/3-torsion root/fibre counts, including identity
geometric_low_torsion_certificate	Actual geometric 2/3-torsion cardinalities4/9 with explicit hypotheses

The proof assumes no rank, basis, Hasse or pairing premise. It uses actual Mathlib points and has 19 pinned formal/config sources. All 30 build/source attempt files and the final gate are retained losslessly. Lean4.35.0 / Mathlib56a00585; only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, waivers or raised elaboration limits. [pcc:unverified]

Native SQL independently reconstructs 7,013,442 short scaling members, 43,330 polynomial point/2/3-torsion counts, 3,929 modular orders and all weighted aggregates. All 25 integrity gates pass. The C++ implementation checks $\#E\cdot P=O$ at all 14,176,080 representative point rows. Higher-r scalar multiplication is computed in C++; SQL checks its aggregates. A/A has 43,605 identical rows and tests determinism on this same host. [pcc:unverified]

The source trace predicate p divides $p+1-\#E$ identifies 197,920 models. Prime-factor and all coprime-divisor embedding-order bounds have zero violations: at most2 for $p>3$ and at most6 including characteristics2/3. Literal equality $k=2$ fails in 197,905 weighted prime-factor cases. Some scanned divisors need not occur as cyclic point orders. This finite count does not prove geometric supersingularity classification. [pcc:unverified]

Source checkpoint b978fdd6 precedes the pilot/main freezes49c8a1f2/6a656748. Protocols, all native outcomes, synthetic controls and exact source/binary/ runtime hashes remain unchanged. A separate build-command clarification records the repository-root source path used by the actual compiler; the original study-relative recipe remains frozen. Every scientific clock is invalid. Native/compiled refinement and independent signed execution remain separate assurance gaps. [pcc:unverified]

Each raw main JSONL is 24,987,542 bytes, above the repository tenMiB Git limit. Original files and the 646,608-byte checked review bundle remain in the approved local checkpoint; hashes and availability are committed in LOCAL_CHECKPOINT.json. Raw data are available in the shared workspace and are not claimed present in a fresh checkout or remotely uploaded. [pcc:unverified]

Primary review: BK1998 Theorem1/proof/Remark1, printed142–143; original PDF ac837b8e. KB unavailable (MCP absent/wrapper127); bounded original-paper, repository and pinned Mathlib fallback. Exact source and verification receipts accompany this report. All-k BK, generic all-prime rank/determinant and the remaining original fourteen-item obligations stay open. [pcc:unverified]